Video Language Teleoperation Models: Scaling Data-Driven Teleoperation with Generalist Policy Representations

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Abstract

Data-driven teleoperation aims to reduce the cognitive burden of controlling high–degree-of-freedom robotic systems by learning mappings from low-dimensional human inputs to robot actions. Existing approaches typically rely on task-specific action prediction, which limits generalization and makes data collection costly in teleoperation settings where target actions are not predefined. In this work, we propose an alternative formulation that leverages pretrained generalist policies to induce a task-conditioned linear control mapping from initial multimodal observations. Rather than continuously predicting actions, the proposed system infers a stable control structure by learning a task-conditioned action covariance, which implicitly defines the geometry and gains of the user-to-robot mapping. We analyze the resulting control law through spectral perturbation analysis, showing that well-separated eigenstructures yield temporally stable and smooth control behavior. We evaluate the approach by fine-tuning the Octo generalist policy on simulated manipulation data across multiple action horizons and assess the expressiveness and robustness of the induced control mappings via deterministic linear reconstruction. Experimental results demonstrate that intermediate training horizons yield improved cross-horizon generalization and that the learned control structure sufficiently covers expert action manifolds. These findings suggest that generalist policy representations can be effectively repurposed to construct stable, low-dimensional teleoperation interfaces without requiring task-specific action supervision.

1 Introduction

Data-driven teleoperation uses machine learning (ML) to assist human operators in controlling robotic systems. It simplifies control interfaces for robots with high degrees of freedom by mapping them to low-dimensional 2D or 3D directional inputs, thereby reducing the operator's cognitive load.

This approach is particularly relevant for assistive teleoperation, where individuals living with disabilities can operate semi-autonomous systems that reportedly enable up to 40% more daily tasks compared to non-users [1].

Such systems typically rely on learning task-specific action mappings. However, this strategy suffers from limited generality because datasets are often goal-specific, which constrains deployment in real-world settings. Recent work has improved generalization by applying visual-language models (VLMs) to action-prediction tasks and by training generalist policies using transformer architectures [2]. In teleoperation, however, target actions are not predefined, which makes data collection for general-purpose models substantially more difficult than in action-prediction settings.

We argue that, given the growing availability of generalist policies, learning a generalist representation and establishing the control mapping once at initialization is a more effective approach. Once the scene is identified, continuous high-level visual or textual feedback is unnecessary, and the controller can instead be specialized using only robot proprioceptive inputs for conditioning.

2 Methodology

In this work, we assume the availability of task definitions $o \in O$ represented as complex observational data. These observations may span multiple modalities, including visual inputs o^{image} and textual descriptions o^{text}. A transformation $h : O \to T$ maps these observations to a task representation $t \in T$, which we model as a vectorized embedding in $\mathbb{R}^n$, where $n \in \mathbb{N}^+$.

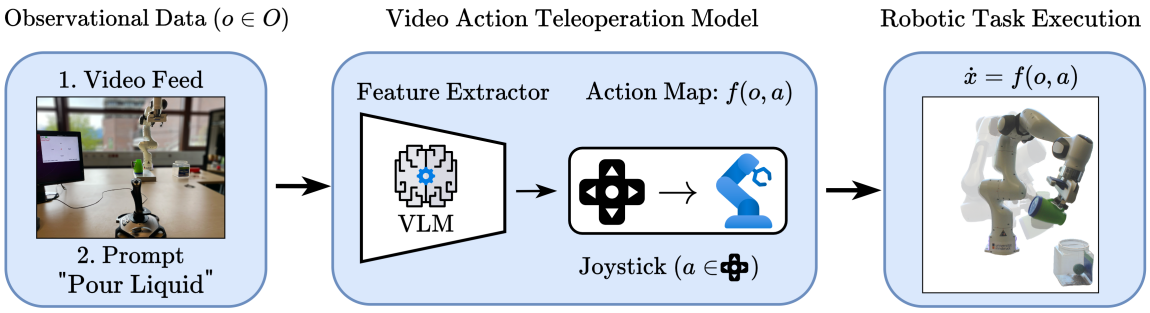


Figure 1: Video–language teleoperation control pipeline. Features extracted by a visual-language model are used as contextual information for downstream control.

Our objective is to exploit a family of mappings H, where each $h \in H$ corresponds to a generalist policy representation, to condition a low-level teleoperation control interface:

$$\dot{x} = f(h(o), x, a), \tag{1}$$

where x denotes the robot state, a the operator input, and f the control dynamics. We assume that, locally, f admits a linear approximation of the form

$$f(h(o), x, a) = W(h(o))a. \tag{2}$$

Here, $W(h(o))$ denotes a task-conditioned linear control mapping induced by the proposed system from the initial observations. This mapping defines how user inputs are transformed into robot actions under the inferred task context.

As a concrete example, consider a teleoperation interface with a 2-DoF bidirectional controller mapped onto an 8-DoF robotic system:

$$\dot{x} = W(h(o))a, \tag{3}$$

where $\dot{x} \in \mathbb{R}^8$ represents robot joint or task-space velocities, $a \in \mathbb{R}^2$ denotes the operator control inputs, and $W(h(o)) \in \mathbb{R}^{8\times 2}$ is the predicted task-specific mapping.

2.1 Theoretical Analysis

For the purpose of this analysis, we treat $H(t, x)$ as a data matrix formed by stacking predicted action vectors produced by the generalist policy, conditioned on task t and state x, over a finite temporal horizon.

We perform an eigenvalue perturbation analysis on the mapping $H(t, x)^\top H(t, x) + \sigma I$. Under small perturbations, standard analytical tools allow us to derive first-order approximations of the resulting spectral changes. This yields a regularized Gram matrix whose spectral structure reflects how control directions are organized and how stable they are under changes in task or state.

When the spectrum is well separated, the dominant control directions encoded by this operator remain stable over time. As a result, the effective coupling between user inputs and robot actions evolves smoothly rather than changing abruptly, which helps preserve a consistent control experience for the user. From this perspective, spectral separation acts as a stabilizing factor for the induced control behavior.

In our implementation, $H(t, x)$ is constructed directly from predicted action sequences produced by the generalist policy over a finite horizon. Consequently, the matrix $H(t, x)^\top H(t, x)$ corresponds to an empirical Gram matrix and can be interpreted as a covariance estimate of the predicted actions. The eigenvalue perturbation analysis therefore operates directly on empirical data:

$$H(t, x)^\top H(t, x) + \sigma I \quad \longleftrightarrow \quad \Sigma_a + \sigma I, \tag{4}$$

where Σ_a denotes the empirical action covariance matrix estimated from the prediction horizon.

Rather than explicitly predicting a linear control matrix, the proposed system induces the task-conditioned control mapping $W(h(o))$ through this predicted action covariance. The spectral structure of Σ_a determines the dominant control directions and their relative gains, while the regularization term improves numerical robustness. In this formulation, both the geometry and the strength of the user-to-robot mapping are encoded implicitly in the covariance, resulting in a stable linear control law that maps user inputs to robot actions in a task-consistent manner.

3 Experiments

In this section, we evaluate whether the task-conditioned control mapping induced by the model adequately covers ground-truth actions and how sensitive it is to different action horizons. We fine-tune the generalist policy model Octo [2] using action horizons of 1, 5, and 15 steps.

To evaluate cross-horizon generalization of the learned control mappings, we perform prediction and reconstruction experiments. Each trained model is evaluated on prediction horizons of 1, 5, and 15 steps, resulting in a grid of training–evaluation horizon pairs shown in Figure 3.

For evaluation, we sample three held-out trajectories from the *aloha_sim* dataset for each experiment. To obtain a reliable estimate of the task-conditioned action covariance, we draw 128 action samples per timestep from the model and compute an empirical covariance matrix. This covariance is then used to derive a low-rank control basis for deterministic linear reconstruction. Predictions are processed in batches of size 64 to improve computational efficiency.

Because the proposed approach induces a linear control mapping through a predicted action covariance rather than directly predicting actions, our evaluation focuses on the representational capacity and temporal stability of the learned control structure rather than pointwise action accuracy.

3.1 Finetuning

The goal of these experiments is to evaluate whether large generalist policy models can be effectively repurposed for teleoperation. To this end, we fine-tune the Octo model on the *aloha_sim* dataset [3], using publicly released simulated manipulation trajectories provided by the Berkeley RAIL group.

Octo follows a transformer-based architecture that processes multimodal observations, including images, proprioceptive state, and language instructions, to produce a latent task representation that is consumed by downstream action heads.

We implement a custom teleoperation action head that predicts a task-conditioned Gaussian distribution over actions. The mean of the distribution is fixed to zero, and the covariance matrix is learned as a function of the task representation. Training is performed by minimizing the negative log-likelihood of ground-truth actions under this predicted distribution. Rather than predicting a separate action at each timestep, the model predicts a single covariance matrix that is shared across a predefined action horizon.

To evaluate the effect of temporal scale, we fine-tune three separate Octo models using action horizons of 1, 5, and 15 steps, each trained for 2000 optimization steps. The training loss curves, shown in Figure 2, suggest similar convergence behavior across all horizons. Overall, the learned action covariance remains well conditioned and trainable across different temporal scales.

3.2 Evaluation

We evaluate the three fine-tuned models across action horizons of 1, 5, and 15 steps. For each training–evaluation horizon pair, we assess the validity, temporal stability, and expressiveness of the induced control mapping using a deterministic linear reconstruction test over three held-out trajectories per experiment.

Given a predicted task-conditioned action covariance $\Sigma_a \in \mathbb{R}^{d\times d}$, we construct a low-rank linear basis by computing its eigendecomposition

$$\Sigma_a = U\Lambda U^\top, \tag{5}$$

and retaining the top-k components,

$$B = U_k\Lambda_k^{1/2} \in \mathbb{R}^{d\times k}, \tag{6}$$

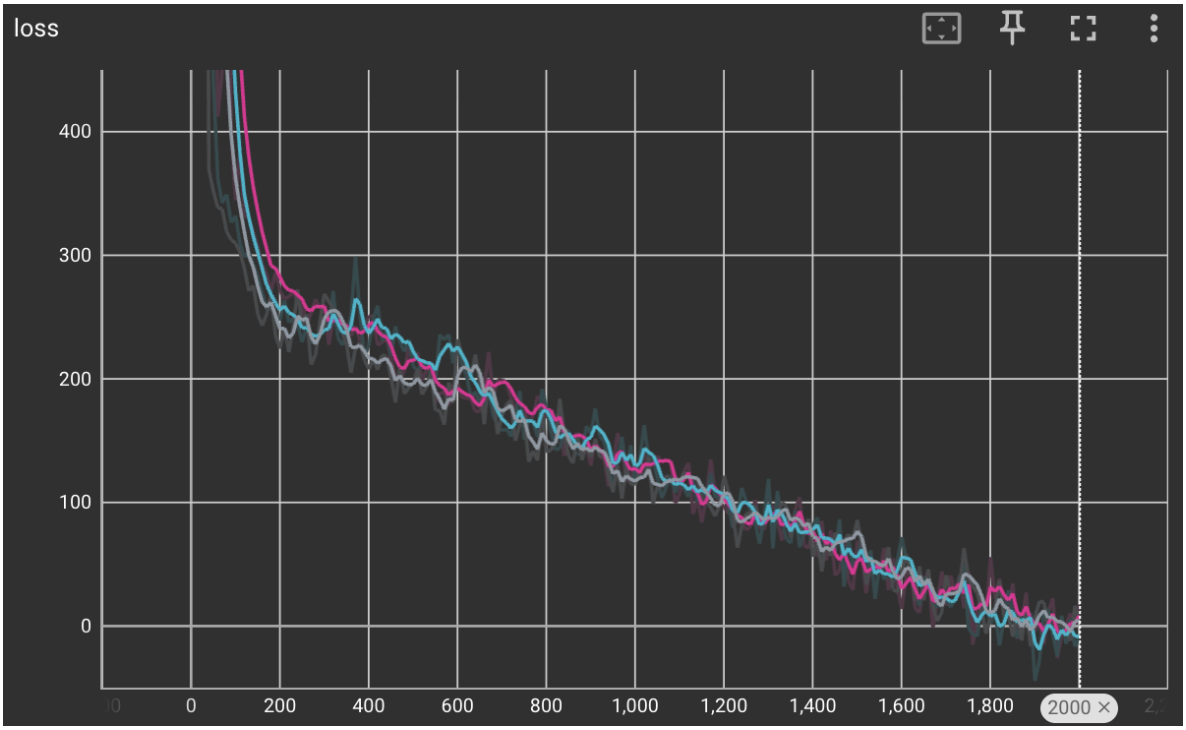


Figure 2: Training loss curves for the teleoperation action head across different action horizons.

where k corresponds to the number of degrees of freedom of the user controller.

For each ground-truth action $a^* \in \mathbb{R}^d$, we solve the following linear least-squares problem:

$$z^* = \arg\min_{z \in \mathbb{R}^k} \|a^* - Bz\|_2^2, \tag{7}$$

and reconstruct the action as

$$\hat{a} = Bz^*. \tag{8}$$

The reconstruction error $\|a^* - \hat{a}\|_2$ measures the extent to which the ground-truth action lies outside the control structure induced by the predicted covariance. Low reconstruction error indicates that the learned control geometry is sufficient to represent expert behavior, while higher error suggests missing or misaligned control directions. Aggregating this error across timesteps and trajectories allows us to evaluate the validity, temporal stability, and expressiveness of the induced control mapping across different action horizons.

3.3 Results

The results in Figure 3 report reconstruction metrics obtained across different training and prediction horizons. Overall, the model trained with an action horizon of 5 demonstrates the most consistent performance, achieving the lowest reconstruction errors for prediction horizons of 1, 5, and 15. In particular, for a prediction horizon of 15, this model yields approximately a threefold improvement in reconstruction error compared to models trained with horizons of 1 and 15.

The observed mean squared error (MSE) values, which lie in the range $[0, 0.2]$, indicate that the proposed approach can induce control mappings that allow operators to closely follow ground-truth trajectories. When training with a horizon of 1, both MSE and mean absolute error (MAE) are significantly higher, suggesting that a single-step objective is insufficient to capture the temporal structure of the task. Conversely, training with a horizon of 15 appears to introduce overfitting, resulting in reduced generalization performance during evaluation.

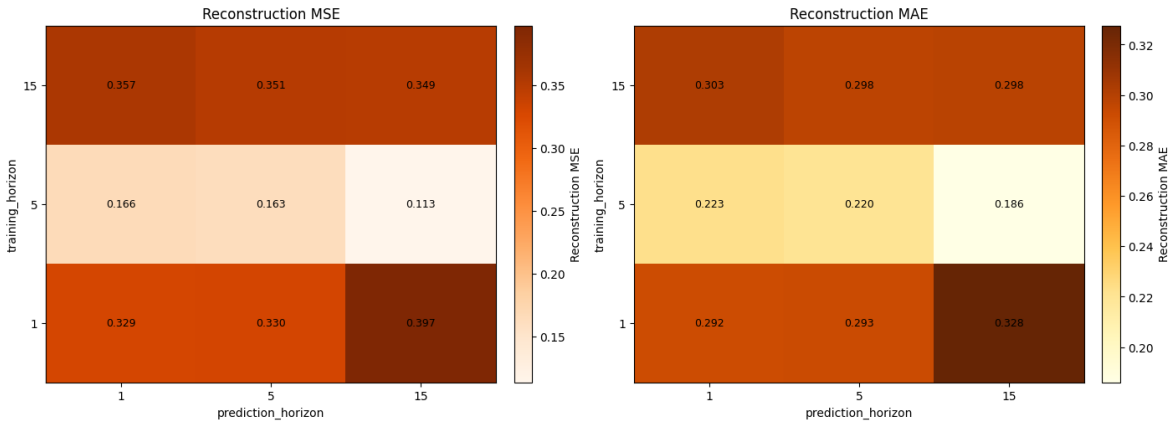


Figure 3: Deterministic linear reconstruction error across training and evaluation horizons.

4 Conclusion & Future Work

In this work, we show that pretrained generalist policy models can be fine-tuned to induce task-specific control mappings from initial visual observations and textual task specifications. The resulting mappings exhibit strong temporal stability and sufficient expressive capacity across multiple action horizons.

Using a publicly available simulated dataset released by the Berkeley RAIL group, we fine-tune the transformer-based Octo model and demonstrate that it induces a meaningful control mapping from a low-dimensional user input space of $\mathbb{R}^2$ to a robot action space of $\mathbb{R}^{14}$. In the best case, this mapping reconstructs ground-truth actions with a deterministic reconstruction MSE of 0.113, indicating effective coverage of the target action manifold.

Potential future directions include qualitative evaluation through user studies and comparisons with alternative control mapping methods. For example, teleoperation experiments involving participants with varying skill levels could be used to assess the practical effectiveness of the induced control mappings, including operator performance, learning dynamics, and subjective satisfaction during task execution.

References

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